# Tata Data Visualisation Job Simulation

Analyze a **UK online retailer's** data to uncover **customer behavior**, **buying patterns**, and **revenue trends** — turning messy records into clear business insight.

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# Loading & Inspecting the Dataset

**541,909 rows · 8 columns**

The dataset is loaded with `pandas` and `numpy`, then checked for the core fields needed for analysis.

- **InvoiceNo**, **StockCode**, **Description**
- **Quantity**, **InvoiceDate**, **UnitPrice**
- **CustomerID**, **Country**

**Top item:** WHITE HANGING HEART T-LIGHT HOLDER

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# Identifying Data Quality Issues

## Missing Descriptions

**1,454** rows with null Description

## Missing CustomerIDs

**135,080** rows without CustomerID

## Negative Quantities

**10,624** rows flagged as returns or adjustments

## Quantity Range

From **−80,995** to **+80,995**; extreme outliers present

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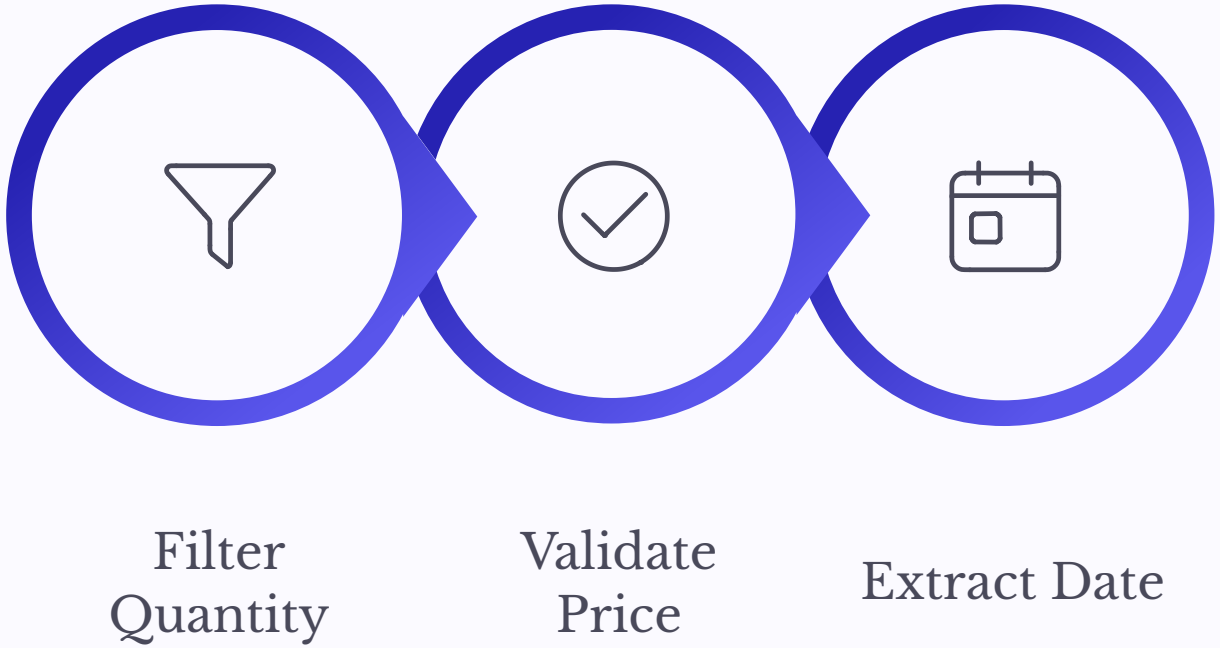
# Cleaning the Data




After filtering, the clean dataset retains **383,058 transactions**. A `Revenue` column is engineered as **Quantity × UnitPrice** to capture financial impact per transaction.

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# Cleaned Dataset: Summary Statistics

| Metric | Quantity | UnitPrice |
|---|---|---|
| Count | **383,058** | **383,058** |
| Mean | **14.39** | **£2.78** |
| Median (50%) | **6** | **£1.79** |
| Max | **80,995** | **£649.50** |

## Key Observations

The gap between the **mean (14.4)** and **median (6)** Quantity points to heavy outliers, likely bulk orders. The **£1.79** median UnitPrice reinforces a low-cost, high-volume retail pattern.

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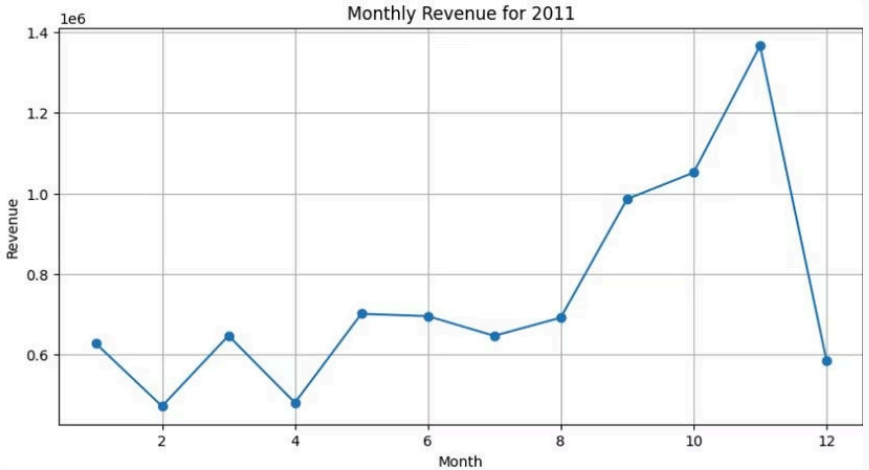
# Monthly Revenue 2011: Strong Q4 Seasonality

Revenue stays near **£0.65M** through mid-year, then **surges from September** to a peak of **£1.38M in November**. December falls to **£0.6M**, likely reflecting incomplete data. The pattern confirms **holiday-driven Q4 demand** as the key revenue window.

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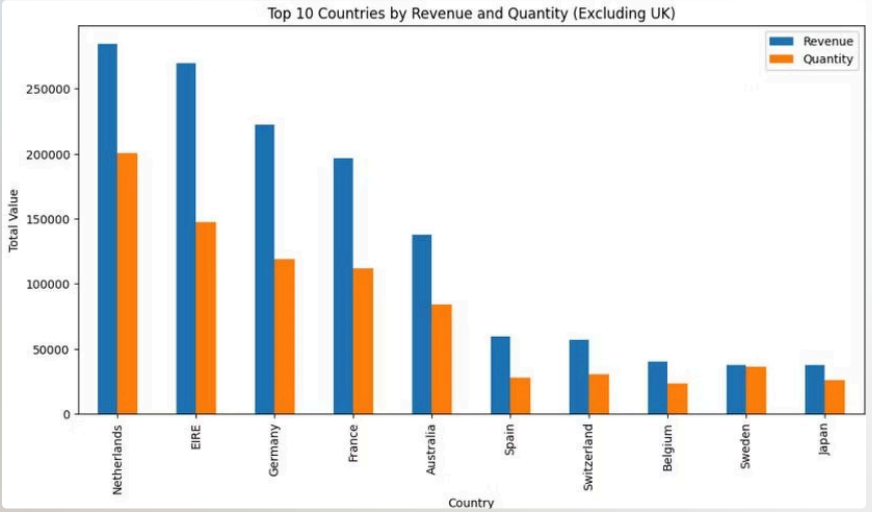
# Top International Markets (Excl. UK)

**Netherlands (~£280K)**, **EIRE/Ireland (~£270K)**, and **Germany (~£220K)** lead international revenue. The Netherlands stands out for its **wholesale-heavy mix**, while **France** and **Australia** remain clear growth opportunities.

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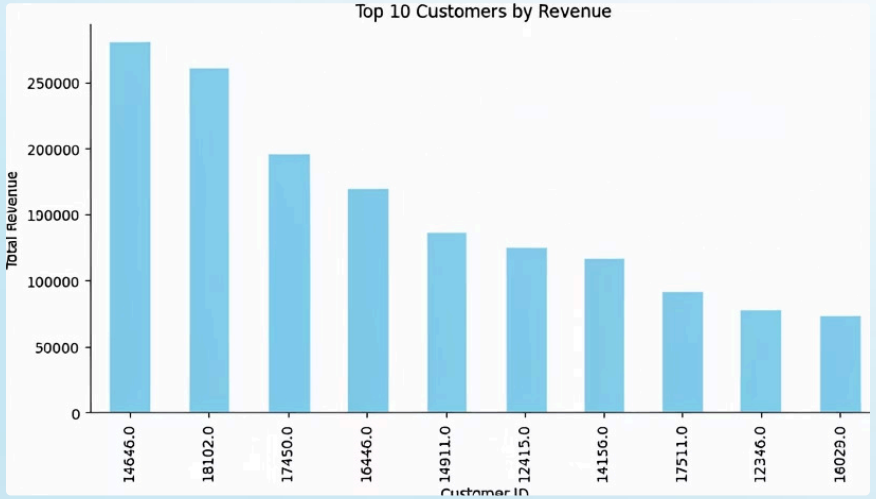


# Top 10 Customers: A "Long Tail" Distribution

Customers **14646** (~£275K) and **18102** (~£260K) dominate revenue, with a sharp drop-off after the top two. This pattern points to a **B2B/wholesale model** where a few accounts drive most sales — and where losing one would create meaningful risk.

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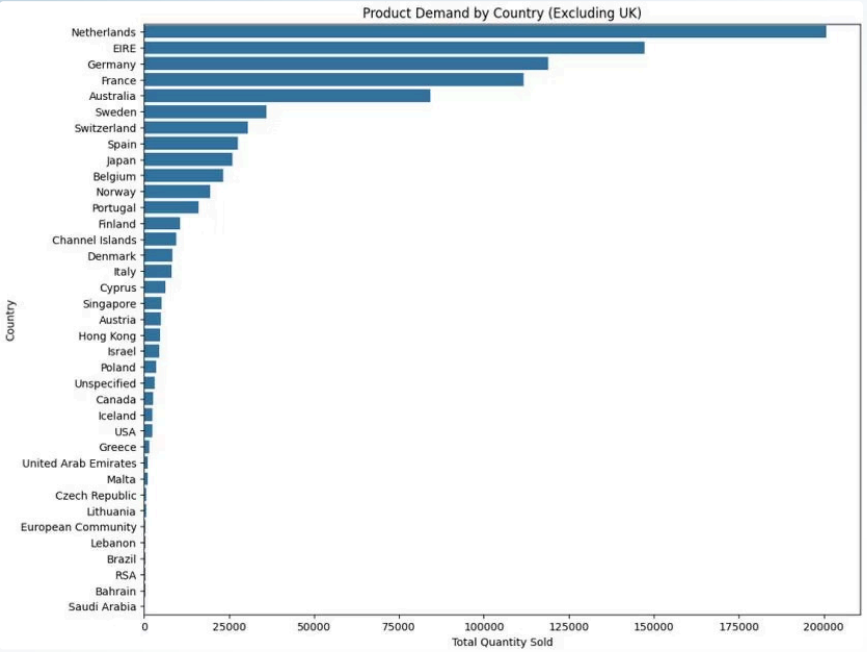
# Product Demand by Country

Demand broadly tracks revenue, with **Netherlands**, **EIRE**, and **Germany** leading. Some markets show **high volume but lower revenue**, pointing to bulk purchases of lower-margin items. Outside the top 5, demand is minimal: **<25,000 units**.

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# Conclusions & Recommendations

### 🏆 VIP Retention

Launch a loyalty programme for the **Top 10 Customer IDs** — losing one account can materially impact revenue.

### 🚚 Logistics Optimisation

Explore **localised warehousing** in the Netherlands, Germany, and Ireland to cut delivery times and costs.

### 📦 Inventory Planning

Build stock by **late August** ahead of the November peak. Prioritise top StockCodes to avoid stockouts.

### 🌍 Market Expansion

Investigate why **France** and **Australia** underperform versus the Netherlands, then target them with focused campaigns.

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